

Caroline Wang

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Research Interests: Human-AI coordination, multi-agent systems, ad hoc teamwork/zero-shot coordination, open-ended learning, reinforcement learning, imitation learning.

Education

University of Texas at Austin

Austin, TX

Final year PhD candidate supervised by Prof. Peter Stone

2026

- **GPA:** 4.0/4.0
- **Selected Coursework:** Learning-based Optimal Control, Reinforcement Learning, Theory of Probability, Neural Networks, Robot Learning, Programming Languages, Graduate Algorithms.

Duke University

Durham, NC

B.S. in Computer Science and Mathematics

2020

Graduation with Highest Distinction.

- **GPA:** 3.93/4.00
- **Selected Coursework:** Machine Learning I/II, Real Analysis I/II, Statistics, Design/Analysis of Algorithms, Higher Dimensional Data Analysis, Probability, Linear Algebra.

Publications

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- **C. Wang**, D. Kasenberg, K. Stachenfeld, P. S. Castro. Discovering Differences in Strategic Behavior Between Humans and LLMs. **ICML 2026**.
 - **C. Wang** et al. JaxAHT: A JAX-based Library for Ad Hoc Teamwork. **MALGAI at ICLR 2026** / Under Submission.
 - **C. Wang**, A. Rahman, J. Cui, Y. Sung, P. Stone. ROTATE: Regret-Driven Open-ended Training for Ad Hoc Teamwork. **arXiv:2505.23686** / Under Submission, 2026.
 - **C. Wang**, A. Rahman, I. Durugkar, E. Liebman, P. Stone. N-Agent Ad Hoc Teamwork. **NeurIPS 2024**.
 - Z. Wang*, **C. Wang***, X. Xiao, Y. Zhu, P. Stone. Building Minimal and Reusable Causal State Abstractions for Reinforcement Learning. **AAAI 2024**.
 - **C. Wang**, I. Durugkar, E. Liebman, P. Stone. DM2: Distributed multi-agent reinforcement learning via distribution matching. **AAAI 2023**.
 - **C. Wang**, G. Warnell, P. Stone. D-Shape: Demonstration-shaped reinforcement learning via goal-conditioning. **AAMAS 2023**.
 - P. MacAlpine, B. Liu, W. Macke, **C. Wang**, P. Stone. UT Austin Villa: RoboCup 2021 3D simulation league competition champions. **Robot World Cup, 2022**.
 - C. Rudin, **C. Wang**, and B. Coker. Broader issues surrounding model transparency in criminal justice risk scoring. **Harvard Data Science Review, 2020**.
 - **C. Wang**, B. Han, B. Patel, F. Mohideen, C. Rudin. In pursuit of interpretable, fair and accurate machine learning for recidivism prediction. **Quantitative Criminology, 2021**.
 - C. Rudin, **C. Wang**, B. Coker. The age of secrecy and unfairness in recidivism prediction. **Harvard Data Science Review, 2020**.

Software

JaxAHT

2025

- Led team of 9 in developing and releasing the first open-source, JAX-based Python library for ad hoc teamwork algorithms and evaluation (<https://larg.github.io/jax-aht/>).

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Professional Experience

Google DeepMind – Student Researcher

June 2025 – Feb 2026

Host: Pablo S. Castro

Project: LLM-powered scientific discovery for human behavior in strategic scenarios.

- Characterized differences in strategic behavior between LLMs and humans using AlphaEvolve, an LLM-powered evolutionary framework for scientific discovery.

Sony AI – Research Intern

Summer 2023

Host: Varun Kompella

Project: Automatically weighing event tables to improve reinforcement learning sample efficiency.

- Developed experience-replay based algorithms to improve sample efficiency of reinforcement learning on Sony game environments.

Data Science for Social Good

Summer 2019

Project: Quantifying traffic dynamics in Greater London to improve air pollution models

- Created open-source, end-to-end system that analyzes London traffic camera feeds using computer vision, and outputs junction-level, near real-time traffic statistics for emissions models.
- Created historical dataset of London traffic statistics. Dataset can be applied to improve emissions models, and/or by local transport authorities to optimize traffic flow/assess effects of roadworks.

Data+

Summer 2018

Supervised by Prof. Jun Yang, Department of Computer Science, Duke University

Project: Automated real-time political fact-checking

- Adapted Transformer model to perform paraphrase matching and entailment detection for political claims.
- Contributed to pipeline for app backend that could analyze an audio source and return related fact-checks from a PostgreSQL database of around 18000 previously checked claims.

Invited Talks

Google DeepMind Reinforcement Learning Reading Group

Oct 2025

ROTATE: Regret-Driven Open-ended Training for Ad Hoc Teamwork.

Multi-agent Reinforcement Learning Reading Group

April 2023

DM²: Distributed multi-agent reinforcement learning via distribution matching.

Cooperative AI Lab

March 2023

DM²: Distributed multi-agent reinforcement learning via distribution matching.

Research Experience

Research Assistant

Fall 2020 – Present

Supervised by Prof. Peter Stone, Department of Computer Science, UT Austin

Project: Regret-driven Open-ended Training for Ad Hoc Teamwork

- Reformulated ad hoc teamwork as an open-ended learning problem and proposed a regret-driven algorithm that achieves state-of-the-art results against out-of-distribution teammates.

Project: Large Language Model Agents in Social Dilemmas

- Led team in participating in the Concordia Challenge at NeurIPS 2024, placing 6th/25.
- Constructed LLM-based agent for dealing with a variety of social scenarios (e.g. negotiation, collective bargaining, alliances) using prompt engineering and chain-of-thought reasoning.

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Project: N-Agent Ad Hoc Teamwork

- Proposed and formulated novel problem of N-agent ad hoc teamwork (NAHT).
- Developed agent modeling based MARL algorithm for NAHT, demonstrating out-of-distribution generalization. Currently exploring multi-agent transformer architectures for decision-making.
- Validated algorithm on StarCraft II and Multi-Agent Particle environments.

Project: Building Minimal and Reusable Causal State Abstractions for Reinforcement Learning

- Developed SAC-based algorithm that learns causal dynamics and reward relationships to generate a state abstraction that is reusable across environments sharing the same dynamics.
- Validated algorithm on RoboSuite environments.

Project: Distributed multi-agent reinforcement learning via distribution matching

- Developed distribution matching algorithm for multi-agent cooperation w/o explicit communication using GAIL and IPPO.
- Validated algorithm on StarCraft II environments.

Project: Deep reinforcement learning for learning a sprint in RoboCup 3DSim

- Parallelized PPO algorithm on a Condor cluster to learn a humanoid robot sprint.
- UTAustinVilla team placed 3rd at RoboCup 3dSim League Competition in 2022.

Project: Combining reinforcement learning with imitation from observation

- Developed novel, model-free reinforcement learning + imitation from observation algorithm.
- Validated algorithm on simulated gridworld environments.

Research Assistant

Fall 2017 – Spring 2020

Supervised by Prof. Cynthia Rudin, Department of Computer Science, Duke University

Project: Interpretable and fair machine learning to predict criminal recidivism

- Determined most appropriate definitions of fairness to apply to the criminal recidivism problem.
- Designed state-of-the-art, interpretable machine learning models to predict criminal recidivism.

Project: Reverse engineering black-box algorithms for criminal recidivism prediction

- Partially reverse-engineered a black-box risk-assessment algorithm (COMPAS) via machine learning and feature analysis.
- Constructed novel criminal recidivism dataset using data sourced from Broward County Clerk's and Sheriff's offices.

Teaching Experience

Teaching Assistant – Reinforcement Learning

Spring 2024

- Held office hours, graded homework, moderated in-class discussion.

Teaching Assistant – Introduction to Ethical AI/Robotics

Fall 2022

- Contributed to course design as a TA for the new course, Introduction to Ethical AI/Robotics, including designing novel homework assignments.
- Held office hours, graded homework, moderated in-class discussion.

Professional Service

Workshop on Multi-Agent Learning and Generative AI (ICLR 2026)

Fall 2025 – Spring 2026

- Organizing committee for the MAL-GAI Workshop.

Editor for Neural Computing & Applications Journal

Fall 2023 – Spring 2024

- Served as editor for the ALA special issue in Neural Computing & Applications (Springer Nature).

Adaptive Learning Agents Workshop (AAMAS 2023 / 24)

Fall 2022 – Spring 2024

- Organizing committee for the ALA 2023 and 2024 Workshops.

Graduate Women and Gender Minorities in Computing (GWGMC)

Spring 2022 – 2025

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- Re-started and co-led the GWGMC student organization at UT Austin.
- Organizer of Women and Gender Minorities in Computing Research Symposium.

Program Committee

Various

- Reviewer for NeurIPS 2026, ICML 2026, AAMAS Journal, ICLR 2026, ICML 2025, NeurIPS 2024, ICLR 2024, GCRL Workshop@NeurIPS 2023, ICML 2023, AAMAS 2023, ALA Workshop@AAMAS 2023, ICRA 2021.

Volunteering

- Conference volunteer at NeurIPS 2025, AAMAS 2024, AAAI 2024, AAMAS 2023, AAAI 2023.
- Served on the admissions committee for the Computer Science Dpt. at UT Austin in Spring 2022.

Awards and Honors

UT Austin Graduate School Fellowship

Summer 2024

- Fellowship support from the Graduate School at UT Austin

UT Austin Dean's Strategic Fellowship

Fall 2020 – Fall 2021

- Fellowship support from the Graduate School at UT Austin.

CRA Outstanding Undergraduate Researcher Award Finalist

2020

- Computing Research Association (CRA) award program recognizes undergraduate students who show outstanding potential in computing research; 23 finalists selected from North America.

Alex Vasilos Memorial Award

2020

- Awarded by Duke Computer Science to two seniors for excellence in computer science research.

Data Science for Social Good Fellowship

Summer 2019

- 38 fellows from 800+ global applicants selected to work on societally impactful data science projects at Imperial College London, UK.

Barry M. Goldwater Scholarship

2019

- Highly prestigious undergraduate STEM research scholarship. 2019 Goldwater Scholars selected from an estimated pool of over 5,000 college students and 443 academic institutions across the United States.

Phi Beta Kappa

Spring 2019

- Invitation-only, national academic honor society for the top 10% of juniors and seniors.

Dean's List w/Distinction

Fall 2019, Fall 2018, Spring 2017

- Awarded to Duke undergraduates with semester GPAs of 4.0/4.0.

Skills

- **Languages:** Python (proficient), C++ (intermediate), R.
- **Tools/Frameworks:** JAX, PyTorch, NumPy, Git, Condor, Slurms.
- **Knowledge:** Multi-agent systems, reinforcement learning, large language models, machine learning, data science, statistics.